Theoretical Justification

Signal-agnostic model validation relies exclusively on background data. Since no unsupervised method can guarantee optimal future anomaly detection for arbitrary anomalies, a necessary assumption is that future anomalies must be deviations along a latent direction that is learnable from typical data. This assumption can be formalized as a monotone likelihood-ratio condition.

1 Assumptions

Assumption 1 (Latent likelihood-ratio recoverability)**.** *Let P_0 denote the typical distribution and P_1 a future anomaly distribution. There exists a scalar latent coordinate*

$$Z = T(X)$$

and a nondecreasing function h such that

$$\frac{dP_1}{dP_0}(x) = h(T(x)).$$

Thus larger values of Z are more anomalous in the Neyman–Pearson sense. In addition, the two typical validation domains are treated as noisy samples of the same latent structure: conditioned on Z, they differ only in the randomness entailed by the sampling process.

Assumption 1 formalizes the standard anomaly-detection premise: a good model learns a coordinate of typicality, and anomalies are detected as departures along that coordinate. Then, the goal of a model validation criterion is not to discover arbitrary unseen anomalies, but to select, among trained candidate detectors, the score that recovers this latent anomaly-relevant coordinate most reliably.

In our setting, trained anomaly detectors already map inputs to scalar scores. We therefore formulate a closed-form version of Assumption 1 directly at the score level, modelling each detector as a noisy measurement of a latent anomaly-relevant coordinate, possibly together with stable nuisance structure.

Assumption 2 (Gaussian score channel with stable nuisance)**.** *Let $\widetilde{S}_\phi$ denote an anomaly score produced by a candidate detector ϕ, assuming that larger values are more anomalous. Under typical data, the two validation samples satisfy*

$$\widetilde{S}_\phi^{(d)} = c_\phi + a_\phi Z + b_\phi U_\phi + \sigma_\phi \varepsilon_\phi^{(d)}, \qquad d \in \{1, 2\},$$

where

$$Z, U_\phi, \varepsilon_\phi^{(1)}, \varepsilon_\phi^{(2)} \overset{\text{ind}}{\sim} \mathcal{N}(0,1), \qquad a_\phi \geq 0, \qquad b_\phi \in \mathbb{R}, \qquad \sigma_\phi \geq 0.$$

The coefficient a_ϕ measures sensitivity to the anomaly-relevant coordinate Z. The coefficient b_ϕ measures sensitivity to a reproducible nuisance coordinate U_ϕ that is stable across validation views *but is not anomaly-relevant. The term* $\sigma_\phi \varepsilon_\phi^{(d)}$ *captures view-specific noise. The standardized scores are:*

$$S_\phi^{(d)} = \frac{\widetilde{S}_\phi^{(d)} - c_\phi}{\sqrt{a_\phi^2 + b_\phi^2 + \sigma_\phi^2}}.$$

Let

$$\rho_\phi^Z = \frac{a_\phi^2}{a_\phi^2 + b_\phi^2 + \sigma_\phi^2}, \qquad \rho_\phi^R = \frac{a_\phi^2 + b_\phi^2}{a_\phi^2 + b_\phi^2 + \sigma_\phi^2}.$$

Here ρ_ϕ^Z *is the anomaly-aligned fraction of score variance, while*

$$\rho_\phi^R = \mathrm{Corr}(S_\phi^{(1)}, S_\phi^{(2)})$$

is the total reproducible fraction of score variance.

As per our initial assumption, future anomalies are positive shifts of the same latent coordinate:

$$Z \mid Y = 1 \sim \mathcal{N}(\delta, 1), \qquad \delta > 0,$$

while U_ϕ *and* $\varepsilon_\phi^{(d)}$ *retain their typical-data laws. Consequently, after the same typical-data normalisation,*

$$S_\phi \mid Y = 0 \sim \mathcal{N}(0, 1), \qquad S_\phi \mid Y = 1 \sim \mathcal{N}(\delta\sqrt{\rho_\phi^Z}, 1).$$

Assumption 2 separates anomaly-aligned reliability ρ_ϕ^Z from total reproducible reliability ρ_ϕ^R. The two coincide only when the stable nuisance component is absent; that is, when $b_\phi = 0$. A detector that is useful for anomaly detection should place reproducible score variation on Z, increasing a_ϕ, rather than on nuisance structure U_ϕ, reducing b_ϕ. A detector can therefore look reproducible on typical data while still being poorly aligned with future anomalies if its reproducible variation is dominated by U_ϕ.

Lemma 1 (TPR is controlled by anomaly-aligned reliability)**.** *Under Assumption 2, for every false-positive rate* $\alpha \in (0, 1)$*,*

$$\mathrm{TPR}_\phi(\alpha) = \Phi\left(\delta\sqrt{\rho_\phi^Z} - \Phi^{-1}(1-\alpha)\right).$$

Hence $\mathrm{TPR}_\phi(\alpha)$ *is strictly increasing in* ρ_ϕ^Z*.*

Proof. Under typical data, $S_\phi \sim \mathcal{N}(0, 1)$, so the FPR-$\alpha$ threshold is $\Phi^{-1}(1-\alpha)$. Under anomalies,

$$S_\phi \mid Y = 1 \sim \mathcal{N}(\delta\sqrt{\rho_\phi^Z}, 1).$$

The stated expression follows by evaluating the anomaly exceedance probability at that threshold. □

2 Reproducible reliability and marginal non-identifiability

We now compare how CAP and marginal stability criteria behave with respect to the reliability parameters in Assumption 2. CAP is evaluated on paired validation scores, so its population objective can vary with the total reproducible reliability ρ_ϕ^R. Wasserstein distance and threshold drift only compare the two validation score histograms. They can measure marginal score drift between validation domains, but they discard the paired dependence that defines ρ_ϕ^R.

2.1 CAP monotonicity

For a binary CAP energy $E(p, y)$, let $p = \nu(s) \in (0, 1)$ denote the normalized score and define the energy contrast

$$\Delta_E(s) = E(\nu(s), 0) - E(\nu(s), 1).$$

The binary Gibbs posterior induced by this energy is

$$q_{\beta,E}(s) = \frac{\exp\{-\beta E(\nu(s), 1)\}}{\exp\{-\beta E(\nu(s), 0)\} + \exp\{-\beta E(\nu(s), 1)\}} = \sigma(\beta \Delta_E(s)),$$
$$m_{\beta,E}(s) = 2q_{\beta,E}(s) - 1 = \tanh\left(\frac{\beta \Delta_E(s)}{2}\right).$$

For the baseline energy $E(p, y) = y(1-p) + (1-y)p$ and the score normalization $\nu(s) = (s+1)/2$, this reduces to the special case $q_{\beta,E}(s) = \sigma(\beta s)$.

Assumption 3 (Monotone binary CAP energy)**.** *The score normalization ν and binary energy E are such that Δ_E is continuously differentiable, nondecreasing, and nonconstant under Gaussian measure:*

$$\mathbb{P}\big(\Delta_E'(G) > 0\big) > 0, \qquad G \sim \mathcal{N}(0, 1).$$

Moreover, the derivatives required below are integrable. Piecewise-smooth energies such as the adaptive ReLU energy satisfy the same argument by a standard smoothing approximation, provided their energy contrast is monotone in the score.

The approximation capacity kernel for a pair (s_1, s_2) is

$$\log\{q_{\beta,E}(s_1)q_{\beta,E}(s_2) + [1 - q_{\beta,E}(s_1)][1 - q_{\beta,E}(s_2)]\} = -\log 2 + \log\{1 + m_{\beta,E}(s_1)m_{\beta,E}(s_2)\}.$$

Since the constant $-\log 2$ is irrelevant for model selection, let L_ϕ be the optimization objective

$$L_\phi(\beta) = \mathbb{E} \log\{1 + m_{\beta,E}(S_\phi^{(1)})m_{\beta,E}(S_\phi^{(2)})\}$$

and $L_\phi^\star$ denote the CAP lift:

$$L_\phi^\star = \sup_{\beta \in [\beta_{\min}, B]} L_\phi(\beta), \qquad 0 < \beta_{\min} \le B.$$

Including $\beta = 0$ gives the uninformative constant posterior; it does not change the selection whenever the optimized CAP value is attained at positive temperature.

Lemma 2 (CAP is monotone in paired Gaussian reliability)**.** *Let (U, V) be standard bivariate Gaussian with correlation $\rho \in [0, 1)$. For every $\beta > 0$,*

$$C_{\beta,E}(\rho) = \mathbb{E}_\rho \log\{1 + m_{\beta,E}(U)m_{\beta,E}(V)\}$$

is strictly increasing in ρ under Assumption 3. Consequently, $\sup_{\beta \in [\beta_{\min}, B]} C_{\beta,E}(\rho)$ is strictly increasing in ρ on $[0, 1]$.

Proof. Let

$$H_{\beta,E}(u, v) = \log\{1 + m_{\beta,E}(u)m_{\beta,E}(v)\}.$$

Price's theorem gives

$$\frac{d}{d\rho}\mathbb{E}_\rho H_{\beta,E}(U, V) = \mathbb{E}_\rho \left[\frac{\partial^2 H_{\beta,E}}{\partial u\, \partial v}(U, V)\right].$$

Since

$$m'_{\beta,E}(t) = \frac{\beta}{2}\Delta'_E(t)\,\mathrm{sech}^2\left(\frac{\beta\Delta_E(t)}{2}\right) \geq 0$$

and this derivative is positive on a set of nonzero Gaussian measure,

$$\frac{\partial^2 H_{\beta,E}}{\partial u\,\partial v}(u,v) = \frac{m'_{\beta,E}(u)m'_{\beta,E}(v)}{\{1+m_{\beta,E}(u)m_{\beta,E}(v)\}^2} \geq 0,$$

with strict positivity after expectation because the bivariate Gaussian density is positive on every product set of positive Lebesgue measure. Therefore $C_{\beta,E}(\rho)$ is strictly increasing for every fixed $\beta > 0$. For the optimized objective, fix $\rho_1 < \rho_2$ and let $\beta_1 \in [\beta_{\min}, B]$ maximize $C_{\beta,E}(\rho_1)$. Then

$$\sup_{\beta\in[\beta_{\min},B]} C_{\beta,E}(\rho_2) \geq C_{\beta_1,E}(\rho_2) > C_{\beta_1,E}(\rho_1) = \sup_{\beta\in[\beta_{\min},B]} C_{\beta,E}(\rho_1).$$

The case $\rho = 1$ follows by continuity. □

Theorem 1 (CAP is monotone in reproducible reliability)**.** *Under Assumptions 2 and 3, the optimized CAP lift $L^\star_\phi$ is strictly increasing in ρ^R_ϕ. Therefore, for any candidate family $\mathcal{A}$,*

$$\arg\max_{\phi\in\mathcal{A}} L^\star_\phi = \arg\max_{\phi\in\mathcal{A}} \rho^R_\phi,$$

with equality of maximizer sets.

Proof. Under Assumption 2, $(S^{(1)}_\phi, S^{(2)}_\phi)$ is a standard bivariate Gaussian pair with correlation ρ^R_ϕ. Therefore

$$L^\star_\phi = \sup_{\beta\in[0,B]} C_\beta(\rho^R_\phi).$$

Lemma 2 implies that this quantity is strictly increasing in ρ^R_ϕ. □

Corollary 1 (CAP–TPR alignment with stable nuisance)**.** *Under Assumption 2, let $\mathcal{A}$ be a candidate family and define the anomaly alignment of the reproducible component by*

$$\pi_\phi = \begin{cases} \dfrac{a_\phi^2}{a_\phi^2+b_\phi^2}, & a_\phi^2+b_\phi^2>0, \\ 0, & a_\phi^2+b_\phi^2=0. \end{cases}$$

Then

$$\rho^Z_\phi = \pi_\phi\rho^R_\phi.$$

Let

$$\mathcal{A}^\star_R = \arg\max_{\phi\in\mathcal{A}} \rho^R_\phi.$$

Every CAP maximizer is TPR-optimal at every false-positive rate $\alpha \in (0,1)$ if and only if

$$\forall\phi^\star \in \mathcal{A}^\star_R,\ \forall\psi\in\mathcal{A}, \qquad \pi_{\phi^\star}\rho^R_{\phi^\star} \geq \pi_\psi\rho^R_\psi. \tag{A}$$

Equivalently, every maximizer of total reproducible reliability must also maximize the anomaly-aligned reliability ρ^Z_ϕ. A simple sufficient condition for (A) is monotone alignment:

$$\rho^R_\phi \geq \rho^R_\psi \quad\Longrightarrow\quad \pi_\phi \geq \pi_\psi \qquad \forall\phi,\psi\in\mathcal{A}. \tag{B}$$

In coefficient form, this states that more reproducible detectors must not have a larger stable-nuisance ratio,

$$\frac{b_\phi^2}{a_\phi^2} \leq \frac{b_\psi^2}{a_\psi^2},$$

with the convention that the ratio is $+\infty$ when $a_\phi = 0$ and $b_\phi \neq 0$.

Proof. The identity $\rho_\phi^Z = \pi_\phi \rho_\phi^R$ follows directly from the definitions. Theorem 1 shows that CAP maximizes ρ_ϕ^R, and Lemma 1 shows that TPR is strictly increasing in ρ_ϕ^Z. Therefore, every CAP maximizer is TPR-optimal if and only if every ρ_ϕ^R-maximizer also maximizes $\rho_\phi^Z = \pi_\phi \rho_\phi^R$, which is exactly (A). Finally, (B) implies (A) because a maximizer of ρ_ϕ^R also has maximal π_ϕ, and hence maximal product $\pi_\phi \rho_\phi^R$. □

Corollary 1 should be read as an alignment condition, not as an assumption that nuisance is absent. CAP rewards reproducible score variation. In the score channel, reproducible variation has an anomaly-relevant part, $a_\phi Z$, and a stable nuisance part, $b_\phi U_\phi$. The ratio

$$\frac{b_\phi^2}{a_\phi^2}$$

measures how much stable nuisance signal the score contains relative to true anomaly signal. Thus, the sufficient condition says that a detector with larger reproducible reliability should not become more stable for the wrong reason. Its reproducible variation should be at least as anomaly-aligned.

The alignment condition follows the contract between training and validation: training must produce scores whose reproducible variation is intended to encode typicality, and CAP then selects the candidate in which that learned structure is most reproducible rather than merely lowest-loss. If the condition fails, CAP can still select a highly reproducible score, but that score may be reproducible because it has learned stable nuisance structure. In a perfect detector, all reproducible variation is anomaly-relevant (i.e. $b_\phi = 0$) and CAP is automatically aligned with TPR.

2.2 Marginal stability non-identifiability

We showed that CAP is aligned with ρ_ϕ^R, and derived conditions under which this alignment also implies alignment with ρ_ϕ^Z and thus with TPR. In this section we will show that marginal stability criteria cannot satisfy this requirement altogether, as ρ_ϕ^R is a property of the joint distribution of the two validation scores, whereas marginal stability criteria only depend on the marginal score distributions $F_{\phi,1}$ and $F_{\phi,2}$.

Definition 1 (Marginal stability criteria)**.** *Let*

$$F_{\phi,d}(t) = \mathbb{P}(S_\phi^{(d)} \leq t), \qquad d \in \{1,2\},$$

be the population CDFs of the score produced by detector ϕ on two typical validation domains. A validation criterion is marginal stability *if there exists a functional Ψ such that*

$$M(\phi) = \Psi(F_{\phi,1}, F_{\phi,2}).$$

Equivalently, if two detectors have the same two marginal score laws, then the criterion assigns them the same value, regardless of their paired dependence or anomaly-detection power.

The Wasserstein and threshold drift baselines are marginal stability criteria, as their population forms are, respectively,

$$M_{W_1}(\phi) = W_1\Big(\mathcal{L}(S_\phi^{(1)}), \mathcal{L}(S_\phi^{(2)})\Big) = \int_0^1 \Big|F_{\phi,1}^{-1}(u) - F_{\phi,2}^{-1}(u)\Big|\, du,$$

and, for target FPR $\alpha \in (0,1)$,

$$M_{\mathrm{thr},\alpha}(\phi) = \Big|\mathbb{P}\Big(S_\phi^{(2)} \geq F_{\phi,1}^{-1}(1-\alpha)\Big) - \alpha\Big| = \Big|1 - F_{\phi,2}\Big(F_{\phi,1}^{-1}(1-\alpha)\Big) - \alpha\Big|,$$

assuming continuity to avoid quantile ties. In contrast, CAP does not belong to this family because its population objective depends on the joint law of $(S_\phi^{(1)}, S_\phi^{(2)})$.

Marginal stability can measure the amount of domain shift between F_1 and F_2, but it cannot identify the paired reproducible reliability ρ_ϕ^R. This is the non-identifiability obstruction.

Theorem 2 (Non-identifiability of reproducible reliability from marginal stability)**.** *Let M be a marginal stability criterion. Fix $\alpha \in (0,1)$, a validation shift $\eta \in \mathbb{R}$, and two reproducible reliability levels*

$$0 \leq \rho_{\phi_-}^R < \rho_{\phi_+}^R \leq 1.$$

There exist two candidate detectors ϕ_- and ϕ_+ whose validation score marginals are identical across candidates,

$$F_{\phi_+,1} = F_{\phi_-,1} = \Phi, \qquad F_{\phi_+,2} = F_{\phi_-,2}, \qquad F_{\phi_+,2}(t) = \Phi(t-\eta),$$

but whose paired reproducible reliabilities are $\rho_{\phi_+}^R$ and $\rho_{\phi_-}^R$. Hence every marginal stability criterion ties the two candidates,

$$M(\phi_+) = M(\phi_-),$$

even though their reproducible reliabilities are different. In particular,

$$M_{W_1}(\phi_+) = M_{W_1}(\phi_-) = |\eta|,$$

and

$$M_{\mathrm{thr},\alpha}(\phi_+) = M_{\mathrm{thr},\alpha}(\phi_-) = \big|1 - \Phi\big(\Phi^{-1}(1-\alpha) - \eta\big) - \alpha\big|.$$

When $\eta = 0$, the common marginal stability value is perfect. When $\eta \neq 0$, the common value is nonzero. In both cases, marginal stability is non-discriminative over ρ_ϕ^R. In the calibrated subcase $\eta = 0$, if

$$\pi_{\phi_+}\rho_{\phi_+}^R > \pi_{\phi_-}\rho_{\phi_-}^R,$$

then Corollary 1 implies that CAP and TPR rank ϕ_+ above ϕ_-, whereas every marginal stability criterion ties the two.

Proof. For each $\rho \in [0,1]$, let

$$G_\rho^{(d)} = \sqrt{\rho}\, Z + \sqrt{1-\rho}\, \varepsilon^{(d)}, \qquad d \in \{1,2\},$$

where $Z, \varepsilon^{(1)}, \varepsilon^{(2)}$ are independent standard Gaussian variables. Then $(G_\rho^{(1)}, G_\rho^{(2)})$ is a standard bivariate Gaussian pair with correlation ρ. Define validation scores by

$$S_\rho^{(1)} = G_\rho^{(1)}, \qquad S_\rho^{(2)} = \eta + G_\rho^{(2)}.$$

Then $S_\rho^{(1)} \sim \mathcal{N}(0,1)$, $S_\rho^{(2)} \sim \mathcal{N}(\eta,1)$, and $\mathrm{Corr}(S_\rho^{(1)}, S_\rho^{(2)}) = \rho$ for every ρ. Instantiate ϕ_- with $\rho = \rho^R_{\phi_-}$ and ϕ_+ with $\rho = \rho^R_{\phi_+}$. The two candidates have the same marginal validation laws, so any marginal stability criterion gives them the same value.

For Wasserstein distance, the two candidate values are both

$$W_1(\mathcal{N}(0,1), \mathcal{N}(\eta,1)) = |\eta|.$$

For threshold drift, the threshold calibrated on the first validation domain is $\Phi^{-1}(1-\alpha)$. Under the second validation domain, the exceedance probability is

$$1 - \Phi\big(\Phi^{-1}(1-\alpha) - \eta\big)\,,$$

which gives the displayed common threshold-drift value. Finally, in the calibrated subcase $\eta = 0$, the displayed strict inequality is the two-model form of $\rho^Z_{\phi_+} > \rho^Z_{\phi_-}$. Since $\rho^R_{\phi_+} > \rho^R_{\phi_-}$, Theorem 1 ranks ϕ_+ above ϕ_-, and Lemma 1 gives the same TPR ordering. $\square$

The same obstruction also explains their unreliability as ranking metrics. If ϕ_- has identical validation marginals and ϕ_+ has a location shift $F_{\phi_+,1} = \Phi$, $F_{\phi_+,2}(t) = \Phi(t-\eta)$ with $\eta > 0$, then smaller-is-better marginal stability criteria prefer ϕ_-. In particular,

$$M_{W_1}(\phi_-) = 0 < M_{W_1}(\phi_+) = \eta, \qquad M_{\mathrm{thr},\alpha}(\phi_-) = 0 < M_{\mathrm{thr},\alpha}(\phi_+),$$

even if $\rho^R_{\phi_-} < \rho^R_{\phi_+}$. When additionally $\pi_{\phi_+}\rho^R_{\phi_+} > \pi_{\phi_-}\rho^R_{\phi_-}$, this means preferring the lower-TPR detector.

3 Gaussian linear anomaly model

We will now illustrate the previous findings with a Gaussian linear anomaly model that has an analytical solution. The model contains an anomaly-relevant latent coordinate, a stable nuisance subspace, and sample-specific noise.

Definition 2 (Gaussian linear model with stable nuisance)**.** *Let $w_\star \in \mathbb{R}^p$ be a unit vector and let $\lambda_Z \in (0,1]$. Let $\Gamma_U \succeq 0$ be a nuisance covariance satisfying*

$$\Gamma_U w_\star = 0, \qquad \lambda_Z w_\star w_\star^\top + \Gamma_U \preceq I.$$

Choose B such that $BB^\top = \Gamma_U$. Under typical data, let the two validation views be

$$X^{(d)} = \sqrt{\lambda_Z}\, Z w_\star + BU + \xi^{(d)}, \qquad d \in \{1,2\},$$

where $Z \sim \mathcal{N}(0,1)$, $U \sim \mathcal{N}(0,I)$, and

$$\xi^{(d)} \overset{\text{ind}}{\sim} \mathcal{N}(0, I - \lambda_Z w_\star w_\star^\top - \Gamma_U)$$

are mutually independent. Future anomalies shift only the anomaly-relevant coordinate:

$$Z \mid Y = 1 \sim \mathcal{N}(\delta, 1), \qquad \delta > 0,$$

while U and $\xi^{(d)}$ keep their typical-data laws.

Corollary 2 (CAP recovery of the likelihood-ratio direction)**.** *In the Gaussian linear model of Definition 2, consider unit-norm linear scores*

$$S_w^{(d)} = w^\top X^{(d)}, \qquad \|w\| = 1, \qquad w^\top w_\star \geq 0.$$

For each w, we can define

$$a_w^2 = \lambda_Z (w^\top w_\star)^2, \qquad b_w^2 = w^\top \Gamma_U w, \qquad \sigma_w^2 = 1 - a_w^2 - b_w^2,$$

which aligns the expression of $S_w^{(d)}$ in terms of Assumption 2, with

$$\rho_w^Z = a_w^2, \qquad \rho_w^R = a_w^2 + b_w^2 = w^\top (\lambda_Z w_\star w_\star^\top + \Gamma_U) w.$$

The likelihood-ratio anomaly direction is $w_\star$, and it is the unique maximizer of $\mathrm{TPR}_w(\alpha)$ for every $\alpha \in (0,1)$. On any candidate family $\mathcal{W}$, CAP maximizers are TPR-optimal within $\mathcal{W}$ whenever

$$\arg\max_{w \in \mathcal{W}} \{a_w^2 + b_w^2\} \subseteq \arg\max_{w \in \mathcal{W}} a_w^2.$$

This is the linear-Gaussian form of the alignment condition in Corollary 1: a score may contain stable nuisance power b_w^2, but nuisance power must not make a less anomaly-aligned direction maximize total reproducible reliability. In particular, over all unit-norm scores with $w^\top w_\star \geq 0$, this condition is implied by the covariance condition

$$\lambda_Z > \|\Gamma_U\|_{\mathrm{op}}.$$

Under this sufficient condition, $w_\star$ is the unique CAP maximizer:

$$\arg\max_{\|w\|=1,\, w^\top w_\star \geq 0} L_w^\star = \{w_\star\} = \arg\max_{\|w\|=1,\, w^\top w_\star \geq 0} \mathrm{TPR}_w(\alpha).$$

Proof. For any unit w, the typical marginal covariance of $X^{(d)}$ is I, so $S_w^{(d)} \sim \mathcal{N}(0,1)$. Moreover,

$$S_w^{(d)} = \sqrt{\lambda_Z}(w^\top w_\star) Z + w^\top B U + w^\top \xi^{(d)}.$$

The nuisance term $w^\top BU$ is a scalar Gaussian with variance $w^\top \Gamma_U w = b_w^2$, and the view-specific term has variance σ_w^2. Thus the linear score has exactly the score-channel form with coefficients a_w, b_w, σ_w. Therefore

$$\rho_w^Z = a_w^2, \qquad \rho_w^R = a_w^2 + b_w^2.$$

Under anomalies, the marginal distribution of X is

$$\mathcal{N}(\delta \sqrt{\lambda_Z}\, w_\star, I).$$

The likelihood ratio is therefore

$$\log \frac{dP_1}{dP_0}(x) = \delta \sqrt{\lambda_Z}\, w_\star^\top x - \frac{\delta^2 \lambda_Z}{2},$$

so the likelihood-ratio direction is $w_\star$. By Lemma 1,

$$\mathrm{TPR}_w(\alpha) = \Phi\Big(\delta \sqrt{\lambda_Z}(w^\top w_\star) - \Phi^{-1}(1-\alpha)\Big),$$

which is uniquely maximized by $w = w_\star$ under the sign constraint $w^\top w_\star \geq 0$. The candidate-family statement follows directly from Corollary 1, because CAP maximizes $\rho_w^R = a_w^2 + b_w^2$, while TPR is strictly increasing in $\rho_w^Z = a_w^2$.

Let

$$\lambda_U = \|\Gamma_U\|_{\mathrm{op}}.$$

Since $\Gamma_U w_\star = 0$,

$$w^\top \Gamma_U w \leq \lambda_U\{1 - (w^\top w_\star)^2\}.$$

Hence, if $\lambda_Z > \lambda_U$,

$$\begin{aligned} \rho_w^R &= \lambda_Z (w^\top w_\star)^2 + w^\top \Gamma_U w \\ &\leq \lambda_U + (\lambda_Z - \lambda_U)(w^\top w_\star)^2 \leq \lambda_Z, \end{aligned}$$

with equality only when $w = w_\star$. Thus $w_\star$ is the unique maximizer of ρ_w^R, and Theorem 1 makes it the unique CAP maximizer. □

The condition $\lambda_Z > \|\Gamma_U\|_{\mathrm{op}}$ is the covariance form of the nuisance-ratio condition. A direction w can have $b_w^2 > 0$, but it cannot be selected by CAP unless its total reproducible power $a_w^2 + b_w^2$ exceeds that of the anomaly direction. The condition prevents this by requiring every stable nuisance direction to be less reproducible than $w_\star$.

Marginal stability behaves differently because it only sees score histograms. In the calibrated version of the above model, unit-norm scores behave as standard gaussians. Therefore Wasserstein distance and threshold drift are zero for every direction w, including directions with $w^\top w_\star = 0$ and hence lower TPR. They do not identify the reproducible covariance that lets CAP select $w_\star$.

The same point appears geometrically in the location-shift construction of Theorem 2. If $P_0 = \mathcal{N}(0, I)$, $P_1 = \mathcal{N}(\delta w_\star, I)$, and the second validation domain has a benign location shift along the anomaly direction,

$$X^{(1)} \sim \mathcal{N}(0, I), \qquad X^{(2)} \sim \mathcal{N}(\eta w_\star, I), \qquad \eta > 0.$$

For any unit-norm direction w with $w^\top w_\star \geq 0$, write

$$c(w) = w^\top w_\star \in [0, 1].$$

Then

$$\mathrm{TPR}_w(\alpha) = \Phi\big(\delta c(w) - \Phi^{-1}(1-\alpha)\big), \qquad M_{W_1}(w) = \eta c(w),$$

and

$$M_{\mathrm{thr},\alpha}(w) = 1 - \Phi\big(\Phi^{-1}(1-\alpha) - \eta c(w)\big) - \alpha,$$

so both marginal stability baselines are strictly increasing in $c(w)$, while TPR is also strictly increasing in $c(w)$. Hence, when the dimension is at least two, every direction in

$$\mathcal{R}_\epsilon = \{w : \|w\| = 1,\ 0 \leq c(w) \leq 1 - \epsilon\}, \qquad \epsilon \in (0, 1),$$

has lower Wasserstein and threshold drift than $w_\star$, even though every such direction has strictly lower TPR than $w_\star$. Thus marginal stability prefers lower-power scores over the true anomaly direction.

4 Finite-sample CAP selection

The practical validation procedure chooses among a finite set of trained candidates using finitely many paired validation samples and a finite numerical search over inverse temperatures. The remaining question is whether the empirical CAP estimate reliably identifies the population CAP maximizer within the searched candidate family. The following result gives the corresponding uniform concentration guarantee.

Theorem 3 (Uniform finite-sample CAP concentration)**.** *Let $\mathcal{A}$ be a finite set of candidate detectors and let $\mathcal{B} = \{\beta_1, \ldots, \beta_K\} \subset [\beta_{\min}, B]$ be the finite temperature grid used to evaluate CAP. For each $\phi \in \mathcal{A}$, let $(S^{(1)}_{\phi,i}, S^{(2)}_{\phi,i})_{i=1}^n$ be i.i.d. validation score pairs from the population paired law of ϕ. Define*

$$\ell_{\phi,\beta}(s_1, s_2) = \log\{q_{\beta,E}(s_1)q_{\beta,E}(s_2) + [1 - q_{\beta,E}(s_1)][1 - q_{\beta,E}(s_2)]\}.$$

Assume the normalized score-energy contrast is bounded on the validation support so that, for all $\phi \in \mathcal{A}$, $\beta \in \mathcal{B}$, and validation pairs,

$$\ell_{\min} \le \ell_{\phi,\beta}(S^{(1)}_\phi, S^{(2)}_\phi) \le \ell_{\max}, \qquad R_\ell = \ell_{\max} - \ell_{\min} < \infty.$$

Let

$$\widehat{L}_\phi(\beta) = \frac{1}{n}\sum_{i=1}^n \ell_{\phi,\beta}(S^{(1)}_{\phi,i}, S^{(2)}_{\phi,i}), \qquad L_\phi(\beta) = \mathbb{E}\,\ell_{\phi,\beta}(S^{(1)}_\phi, S^{(2)}_\phi),$$

and define the grid-optimized objectives

$$\widehat{L}^\star_{\phi,\mathcal{B}} = \max_{\beta\in\mathcal{B}} \widehat{L}_\phi(\beta), \qquad L^\star_{\phi,\mathcal{B}} = \max_{\beta\in\mathcal{B}} L_\phi(\beta).$$

Then, with probability at least $1 - \delta$,

$$\max_{\phi\in\mathcal{A}} \left|\widehat{L}^\star_{\phi,\mathcal{B}} - L^\star_{\phi,\mathcal{B}}\right| \le R_\ell\sqrt{\frac{\log(2|\mathcal{A}|K/\delta)}{2n}}.$$

Consequently, if $\widehat{\phi} \in \arg\max_{\phi\in\mathcal{A}} \widehat{L}^\star_{\phi,\mathcal{B}}$, then

$$L^\star_{\widehat{\phi},\mathcal{B}} \ge \max_{\phi\in\mathcal{A}} L^\star_{\phi,\mathcal{B}} - 2R_\ell\sqrt{\frac{\log(2|\mathcal{A}|K/\delta)}{2n}}.$$

If the population grid-CAP maximizer is unique and its gap over the second-best candidate exceeds twice the displayed deviation bound, then the empirical CAP maximizer equals the population grid-CAP maximizer.

Proof. For fixed (ϕ, β), Hoeffding's inequality gives

$$\mathbb{P}\Big(|\widehat{L}_\phi(\beta) - L_\phi(\beta)| > t\Big) \le 2\exp\left(-\frac{2nt^2}{R_\ell^2}\right).$$

A union bound over the $|\mathcal{A}|K$ candidate-temperature pairs gives, with probability at least $1 - \delta$,

$$\max_{\phi\in\mathcal{A},\beta\in\mathcal{B}} |\widehat{L}_\phi(\beta) - L_\phi(\beta)| \le R_\ell\sqrt{\frac{\log(2|\mathcal{A}|K/\delta)}{2n}} \equiv \epsilon_n.$$

Taking maxima over β is 1-Lipschitz with respect to the sup norm, so

$$\left| \max_{\beta \in \mathcal{B}} \widehat{L}_\phi(\beta) - \max_{\beta \in \mathcal{B}} L_\phi(\beta) \right| \leq \epsilon_n$$

simultaneously for every ϕ. This proves the first display. If $\widehat{\phi}$ maximizes the empirical grid objective and $\phi^\star$ maximizes the population grid objective, then

$$L^\star_{\widehat{\phi},\mathcal{B}} \geq \widehat{L}^\star_{\widehat{\phi},\mathcal{B}} - \epsilon_n \geq \widehat{L}^\star_{\phi^\star,\mathcal{B}} - \epsilon_n \geq L^\star_{\phi^\star,\mathcal{B}} - 2\epsilon_n.$$

The exact-selection statement follows by comparing this inequality with the population gap. □

Writing

$$\epsilon_n = R_\ell \sqrt{\frac{\log(2|\mathcal{A}|K/\delta)}{2n}},$$

Theorem 3 separates estimation error from the population alignment question. Empirical CAP selection is a statistically consistent proxy for population CAP selection over a fixed search set, and the cost of searching many candidates and temperatures enters only through $\log(|\mathcal{A}|K)$. If two candidates have a population CAP gap smaller than $2\epsilon_n$, finite validation noise can change their empirical ordering. If the population gap is larger than $2\epsilon_n$, the empirical maximizer recovers the population maximizer with probability at least $1 - \delta$.

Remark 1. *In practice, CAP optimizes β over a compact continuous interval $[\beta_{\min}, B]$, rather than over a fixed grid. The same guarantee extends by a standard covering argument. If $\ell_{\phi,\beta}$ is uniformly L_β-Lipschitz in β, choose an η-net $\mathcal{B}_\eta$ of $[\beta_{\min}, B]$, with $|\mathcal{B}_\eta| \leq \lceil (B - \beta_{\min})/\eta \rceil + 1$. Applying the theorem on $\mathcal{B}_\eta$ gives, uniformly over the continuous interval, an additional approximation error at most $2L_\beta\eta$:*

$$\sup_{\phi \in \mathcal{A}} \left| \sup_{\beta \in [\beta_{\min}, B]} \widehat{L}_\phi(\beta) - \sup_{\beta \in [\beta_{\min}, B]} L_\phi(\beta) \right| \leq R_\ell \sqrt{\frac{\log(2|\mathcal{A}||\mathcal{B}_\eta|/\delta)}{2n}} + 2L_\beta\eta.$$

Thus continuous optimization changes the result only by the discretization term $2L_\beta\eta$, which can be made arbitrarily small on compact temperature intervals.